Chilli bacterial and viral diseases

Bacterial diseases

1. **Bacterial spot** - *Xanthomonas campestris* pv. *vesicatoria*
2. Bacterial wilt - *Ralstonia solanacearum*

Viral diseases

1. **Leaf curl** - *Chilli Leaf Curl Virus* (ChiLCV)

2. **Mosaic** - *Potato Virus Y* (PVY) and its strains, *Cucumber Mosaic Virus* (CMV), *Tobacco Mosaic Virus* (TMV), *Tobacco Etch Virus*, *Pepper Vein Banding Virus*, *Pepper Mottle Virus* and *Pepper Severe Mosaic Virus*

Bacterial spot - *Xanthomonas campestris* pv. *vesicatoria*

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Bacterial spot on capsicum

Capsicum—leaf spots

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Bacterial spot - *Xanthomonas campestris* pv. *vesicatoria*

Bacterial spot - *Xanthomonas campestris* pv. *vesicatoria*

- The disease affects bell pepper/ chillies and tomato. Bacterial spot of capsicum was first reported from Florida, USA by Sherbakoff (1918), but in India, disease was observed for the first time in Pune during 1948.
- Common in areas with significant precipitation and temperature during growth and fruiting stage.

Symptoms:

- On leaves, initially the lesions are water-soaked, circular or irregular that becomes necrotic with brown centers and thin chlorotic borders

Bacterial spot - *Xanthomonas campestris* pv. *vesicatoria*

Symptoms contd..

- These lesions are generally sunken on the upper surface of the leaves and slightly raised on the bottom.
- In favourable environmental conditions, these spots coalesce and give a blighted appearance and such leaves turn yellow and fall down prematurely.
- Lesions may enlarge to 10mm in dia, spots may coalesce under favorable conditions with blighted appearance.

Bacterial spot - *Xanthomonas campestris* pv. *vesicatoria*

Symptoms contd…

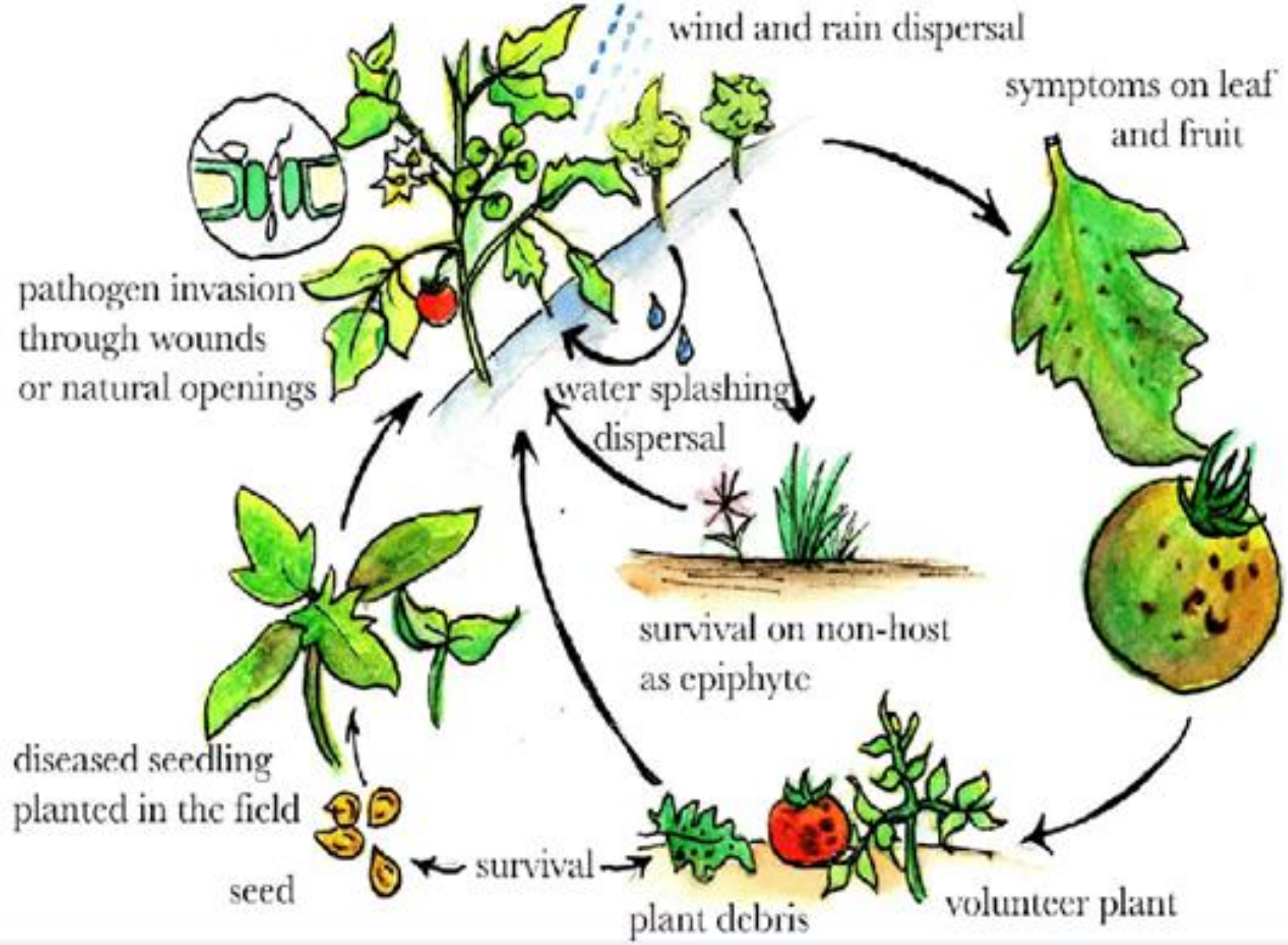
- **Lesions on the stem** are narrow, elongated and raised expanding upto 6 mm
- **Lesions on fruits** initially begin as green spots, which enlarge and later become brown in colour
- These spots are raised with a cracked, roughened wart like appearance and upto 5 mm in dia
- During periods of high humidity, fruit around the lesions may start rotting.
- Similar spots on floral parts, cotyledons and petioles.

Bacterial spot - *Xanthomonas campestris* pv. *vesicatoria*

Pathogen:

- The bacterium responsible for this disease is *Xanthomonas campestris* pv. *vesicatoria*
- The bacterium is motile, strictly aerobic, Gram -ve, rod shaped with round ends, non-spore forming and possesses a single polar flagellum (Monotrichous).
- Bacteria occur singly but occasionally in pairs and rarely in short chains.
- Its colonies on nutrient agar or on specific medium yeast dextrose agar (YDA) are lemon yellow, slightly convex, wet and shining.

Bacterial leaf spot: *Xanthomonas campestris pv. vesicatoria* (Pammel) Dowson



Bacterial spot - *Xanthomonas campestris* pv. *vesicatoria*

Disease cycle and epidemiology:

- The **bacterium persists in infected seeds and plant debris.**
- Moderate temperatures along with high precipitation and relative humidity favour disease development.
- Infection of the plants can take place at a wide range of temperature i.e. 15 to 35°C.
- The optimum temperature lies between 22 to 34°C and maximum disease develops in between July and September.
- The bacterium is disseminated within a field by wind driven rain droplets, clipping of transplants and aerosols.

Bacterial spot - *Xanthomonas campestris* pv. *vesicatoria*

Management:

- Since the disease development requires high precipitation and high relative humidity, fields should be well drained, free of low-lying areas to minimize water-logged conditions.
- The field should be rotated with non-solanaceous crops and bell pepper/ chillies should not be rotated with tomato.
- Use of disease free seeds also reduce the immediate availability of primary inoculum and ensures disease free transplants.
- Seed treatment with hot water at 50^0C/25minutes

Bacterial spot - *Xanthomonas campestris* pv. *vesicatoria*

Management contd…

- **Disinfecting the seed with sodium hypochlorite 1.3% by dipping for one minute followed by shade drying**
- Sprinkler or overhead irrigation of the field should be avoided to keep the disease under check.
- Treat the seeds by dipping in Streptocycline (100 ppm) for 30 minutes.
- Spray the crop with Streptocycline (100 ppm) with the initiation of the disease.

Bacterial wilt - *Ralstonia solanacearum*

Sudden wilting, no yellowing

Healthy (left) and infected (right) plants; notice the internal discoloration of infected vascular tissue and external rotting of roots

Bacterial ooze from infected tomato stem (right); the left stem is healthy

Bacterial wilt - *Ralstonia solanacearum*

Figure 8. Internal stem symptoms of bacterial wilt.

Bacterial wilt - *Ralstonia solanacearum*

Management

Similar to tomato and brinjal

Cultivation of wilt resistant chilli cvs. Dark Purple, Long Red and Pant C-l.

Chilli Leaf Curl -*Chilli leaf curl virus* (ChiLCV)

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- **Other vernacular name - Churda murda**
- The disease causes heavy losses in yield and quality of fruits.
- If the plants are infected within 20-25 days after transplanting, the loss in yield goes up to 80-90%.
- However, in case of later infections, the loss is comparatively less

Chilli Leaf Curl -*Chilli leaf curl virus* (ChiLCV)

Symptoms:

- The symptom of the disease appears as curling of the leaves followed by reduction in their size which later turn pale yellow in colour.
- The older leaves become leathery and brittle.
- Nodes and internodes are significantly reduced in size.
- Infected plants assume a bushy appearance, with severely stunted growth, look pale and produce more lateral branches.
- Fruits formation in susceptible cultivars is rudimentary and distorted.

Chilli Leaf Curl -*Chilli leaf curl virus* (ChiLCV)

Genus: ***Begomovirus*** **of Geminiviridae**

Single-stranded DNA virus.

- The virus consists of geminate isometric particles measuring 18 to 20 nm in diameter.

Chilli Leaf Curl -*Chilli leaf curl virus* (ChiLCV)

Epidemiology:

- Field-spread occurs through **whitefly (*Bemisia tabaci*).**
- This disease increases with increases in temperature and relative humidity.
- **Warm and dry weather favours disease spread**.
- In southern India, disease epidemics are more common during March-June
- whereas under north Indian conditions, epidemics occur from June to October.
- The virus mainly perpetuates on weed hosts.

Chilli Leaf Curl -*Chilli leaf curl virus* (ChiLCV)

Epidemiology contd…

- Alternate hosts – **tomato, papaya, sesamum ,bean, petunia and sugarbeet**

- During **wet weather disease spread was slow** due to reduced vector activity

- Disease epidemics depend on vector availability, source of inoculum, vector activity, vector population and prevailing environmental conditions

Chilli Leaf Curl -*Chilli leaf curl virus* (ChiLCV)

Management :

Cultural and mechanical control

- Growing the nursery under **nylon-net** cover (50 mesh)
- **Eradication of early-infected plants and weed hosts** from the field
- **Two rows of border crop** with maize, jowar, or bajra reduce disease-spread
- Installation of **yellow sticky traps**

Chilli Leaf Curl -*Chilli leaf curl virus* (ChiLCV)

- **Management : Chemical control**
  Soil-application of Furadon @ 1.5 Kg ai/ha at the time of sowing.
- Spray seedlings with Acephate (0.15%) or Monocrotophos (0.1 %) prior to transplanting.
- Spray insecticides like Monocrotophos (0.15%), Acephate (0.15%) or Hostothion (0.1 %) , Imidacloprid, acetamaprid, cypermethrin at fortnightly intervals after transplanting, until the flowering stage.
- Chemical spray, followed by neem seed kernel extract (2%), is also effective in rotation with insecticides.

- Growing of resistant varieties **Pusa sadabahar, Pusa jwala, Jawahar mirchi-2, Pant C-2**

Chilli Leaf Curl -*Chilli leaf curl virus* (ChiLCV)

Management :

Biological control

- Use of biocontrol agent like ***Encarsia bimaculata, Encarsia formosa***
- The greenhouse **white fly, *Trialeurodes vaporarium* gave effective control of *Bemisia tabaci*.**
- Biopesticides based on microbials such as *Beauveria bassiana* (effective on nymphs and adults) or *Isaria fumosoroseus*.

Chilli mosaic complex

Pathogens

Potato Virus Y (PVY) and its strains, *Cucumber Mosaic Virus* (CMV), *Tobacco Mosaic Virus* (TMV), *Tobacco Etch Virus*, *Pepper Vein Banding Virus*, *Pepper Mottle Virus* and *Pepper Severe Mosaic Virus*

Chilli mosaic complex

- **Different types of mosaic pattern are produced depending upon the type of virus or virus (s) involved.**

- Potato virus Y and its strains generally induce vein and stem necrosis.

- Pepper veinal mottle virus causes mosaic, mottling, distortion and filiform of the leaves leading to bushy appearance of the infected plants

- Cucumber mosaic virus produces **mosaic, mottling reduction in leaf lamina and filiformy**.

Chilli mosaic and leaf distortion disease
(*Cucumber Mosaic Virus)*

Chilli mosaic and leaf distortion disease (CMV)

Symptoms:

- Symptoms are extremely variable depending on the variety/ hybrid and on weather conditions.
- Common symptoms are: initial chlorotic lesions, followed by light- and dark-green areas appearing as mosaic.
- At later stages, newly emerging leaves distort or transform into filiform or thin, elongated leaves.
- In severe cases, reduction in leaf lamina and filiformy. plants get stunted and produce less number of fruits.
- Fruits from infected plants are hard and brittle, full of seeds.
- Yellow streaks may also appear on fruits.

Chilli mosaic and leaf distortion disease
(*Cucumber Mosaic Virus)*

Pathogen

The virus is an isometric particle of 28 nm size, has a genome **of single-stranded multipartite RNA** and belongs to the **Cucumovirus group**

Epidemiology:

The disease spreads by **mechanical contact and also through seed and insect-vector aphids** (*Aphis gossypii*, *A. craccivora*, *Myzus persicae*).

Chilli mosaic and leaf distortion disease
(*Cucumber Mosaic Virus)*

Management:

Nursery stage

1. Growing the nursery under nylon-net cover (40-50 mesh).
2. Soil application of systemic insecticides like Furadon @ 1.0 Kg a.i / ha at the time of sowing seeds in the nursery bed.
3. Prior to transplantation, seedlings should be sprayed with Acephate (0.15%) or Monocrotophos @ (0.15 %) or Dimethoate (0.2 %).

Chilli mosaic and leaf distortion disease
(*Cucumber Mosaic Virus)*

Management: Main field

1. Sowing border crops like maize/ bajra/jowar, 15 days before transplanting the main crop.
2. Removal of early-infected plants from the field.
3. Spraying Acephate (0.15 %) or Hostothion (0.1 %) or Imidacloprid (0.05%); these chemicals should be used alternatively and should not be repeated continuously.
4. Spraying neem seed kernel extract (2 %) + sticker, 15 days after transplanting until fruit formation, at 10 days intervals.
5. Mulching with silver or black colored polythene mulch-sheet.
6. Growing virus tolerant varieties.

Chilli mild mottle and mosaic disease- (*Pepper Mild Mottle Virus* and *Tobacco Mosaic Virus*)

Chilli mild mottle and mosaic disease- (*Pepper Mild Mottle Virus* and *Tobacco Mosaic Virus*)

Symptoms:

The disease is characterized by mild mottling with light green areas, interveinal chlorosis, mosaic and distortion of fruits, in some varieties/ hybrids the half portion of leaf will be pale yellow color.

The disease generally does not affect the growth of the plant.

This virus belongs to tobamo virus group, which are having a rod shaped particles measuring 670 to 710 nm in length with 18 nm diameter.

They have single stranded RNA genome.

Chilli mild mottle and mosaic disease- (*Pepper Mild Mottle Virus* and *Tobacco Mosaic Virus*)

Epidemiology:

These viruses are externally seed borne and spread in the field mainly due to contact and injury to the crop during cultural operations in the crop.

The virus mainly survives in the outer seed coat and seldom in the endosperm.

Chilli mild mottle and mosaic disease- (*Pepper Mild Mottle Virus* and *Tobacco Mosaic Virus)*

Management

1. The main preventive measure to protect the viruses is to use virus free seed.
2. Seed borne infection can be prevented by soaking seeds in **Calcium hypochlorite (5 %) for 15 minutes or Trisodium phosphate (10%) solution for 20-30 min and drying**.
3. Raising seedlings under nylon nets (50 mesh).
4. Roguing out infected plants and weed-hosts.
5. Growing two rows of the border crop with maize /jowar/ bajra 15 days before transplanting chilli.